

# BB667: Computational Physiology

19/02/2026

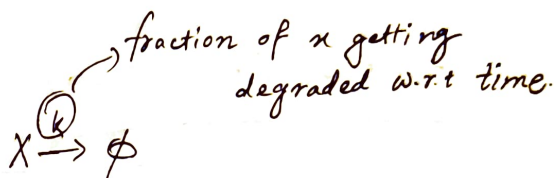
## Building ODE Models for biological Processes

### Law of Mass Action

Rate of an elementary reaction is directly proportional to the Product of the activities  $\frac{\text{or}}{\text{or}}$  'Concentrations of the Reactants'.

Reaction Rate Constant  $K$

Degradation



Rate of change of  $x \propto [x]$   
 $\propto K$

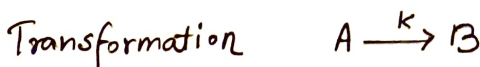
$$\frac{dx}{dt} = (-)K[x]$$

$$\Rightarrow x = x_0 e^{-Kt}$$



$$\frac{dx}{dt} = K$$

$$\Rightarrow x = x_0 + kt$$



$$\frac{dA}{dt} = -KA$$

$$\frac{dB}{dt} = +KA$$

## Dissociation Reaction



$$\frac{dz}{dt} = -kz$$

$$\frac{dX}{dt} = +kz$$

$$\frac{dY}{dt} = +kz$$

$$\Rightarrow z = z_0 e^{-kt}$$

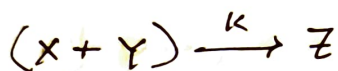
$$\frac{dX}{dt} = kz_0 e^{-kt}$$

$$\Rightarrow \int dX = \int kz_0 e^{-kt} dt$$

$$\Rightarrow X = \frac{kz_0 e^{-kt}}{-k} + C \quad ??$$



## Association Reaction

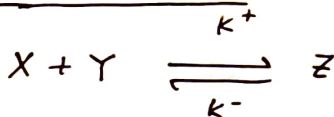


$$\frac{dz}{dt} = (+)kXY$$

$$\frac{dX}{dt} = (-)kXY$$

$$\frac{dY}{dt} = (-)kXY$$

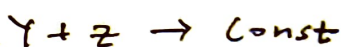
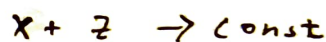
## Bidirectional Reaction



$$\frac{dX}{dt} = k^- z - k^+ XY$$

$$\frac{dY}{dt} = k^- z - k^+ XY$$

$$\frac{dz}{dt} = k^+ XY - k^- z$$

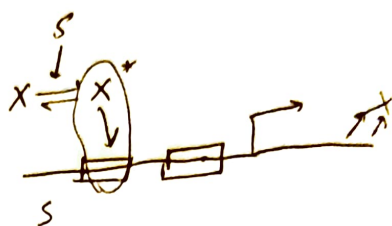
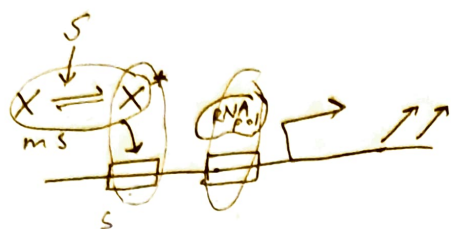
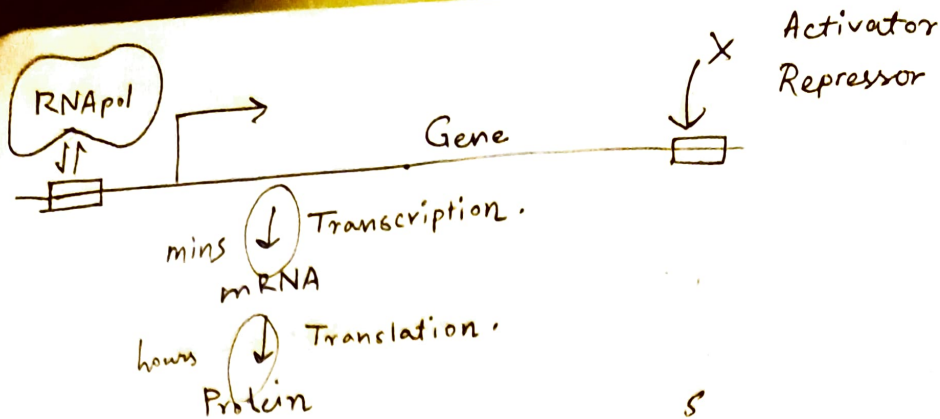


$$X = X_0 - z$$

$$Y = Y_0 - z$$

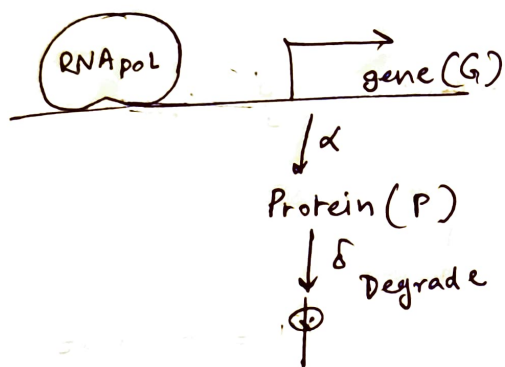
$$\frac{dz}{dt} = \frac{k^+ (X_0 - z)(Y_0 - z)}{-k^- z}$$

Conservation Law  $\rightarrow$  Total Mass / Atoms remain constant.



Time Scale Separation

# Modelling



generally  
one copy

$$\frac{dP}{dt} = + \alpha G - \delta P$$

$$= \alpha - \delta P$$

Case ①  $\frac{dP}{dt} = -\delta P \Rightarrow dP = -\delta P dt$

$\Rightarrow P = P_0 e^{-\delta t}$

$\Rightarrow P = P_0 e^{-\delta t}$

at  $t_h$

$P = \frac{P_0}{2}$

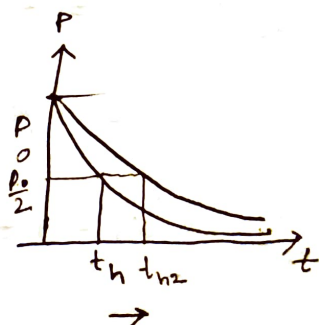
$\frac{P_0}{2} = P_0 e^{-\delta t_h}$

$\Rightarrow e^{-\delta t_h} = \frac{1}{2}$

$\Rightarrow e^{\delta t_h} = 2$

$\Rightarrow \delta t_h = \ln 2 \Rightarrow t_h = \ln 2 / \delta$

$\Rightarrow t_h = 0.693 / \delta$



$t_h \propto \frac{1}{\delta}$

Steady State of P

$$\frac{dP}{dt} = \alpha - \delta P_s = 0$$
$$\Rightarrow P_s = \frac{\alpha}{\delta}$$

$$P_0 = \frac{\alpha}{\delta}$$

$$\therefore P = P_0 e^{-\frac{\delta}{\alpha} t}$$
$$\Rightarrow P = \frac{\alpha}{\delta} e^{-\delta t}$$

Case II

$$t=0, P_0=0$$

$$IF = e^{\int \delta dt} = e^{\delta t}$$

$$\frac{dP}{dt} = \alpha - \delta P$$

$$\Rightarrow \frac{dP}{dt} + \delta P = \alpha$$

$$\Rightarrow e^{\delta t} \frac{dP}{dt} + e^{\delta t} \delta P = \alpha e^{\delta t}$$

$$\Rightarrow \frac{d}{dt} (P e^{\delta t}) = \alpha e^{\delta t}$$

$$\Rightarrow P e^{\delta t} = \int \alpha e^{\delta t} dt$$

$$\Rightarrow P e^{\delta t} = \frac{\alpha}{\delta} e^{\delta t} + C$$

$$t=0 \Rightarrow P=0 \Rightarrow 0 = \frac{\alpha}{\delta} + C \Rightarrow C = -\frac{\alpha}{\delta}$$

$$\Rightarrow P e^{\delta t} = \frac{\alpha}{\delta} e^{\delta t} - \frac{\alpha}{\delta}$$

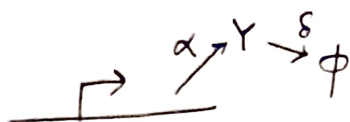
$$\Rightarrow P_t = \frac{\alpha}{\delta} - \frac{\alpha}{\delta} e^{-\delta t}$$

$$\Rightarrow P_t = \frac{\alpha}{\delta} (1 - e^{-\delta t})$$

# BB667: Computational Physiology

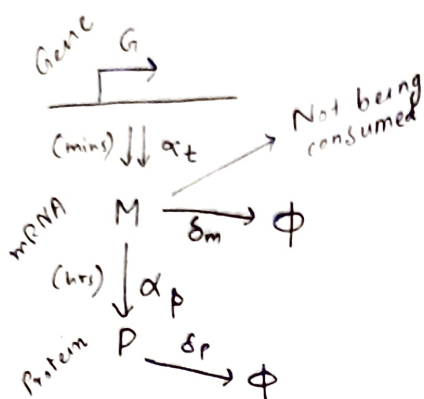
02/03/2026

Gene Expression



$$\frac{dY}{dt} = \alpha G - \delta Y$$

$$\frac{dY}{dt} = \alpha - \delta Y$$



$$\frac{dM}{dt} = \alpha_t G - \delta_m M - \alpha_p M = 0$$

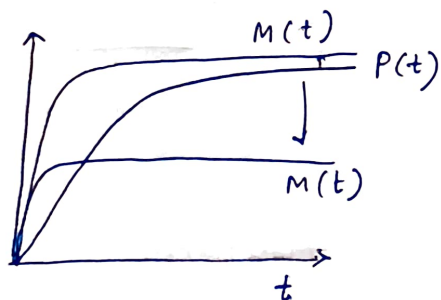
$$\frac{dP}{dt} = \alpha_p M - \delta_p P = 0$$

take  $G = 1$

Steady State

$$M_{ss} = \frac{\alpha_m G}{\delta_m} = \frac{\alpha_m}{\delta_m}$$

$$P_{ss} = \frac{\alpha_p M_{ss}}{\delta_p} = \left( \frac{\alpha_p}{\delta_p} \right) \frac{\alpha_m}{\delta_m}$$



$M(t)$  rises quicker than  $P(t)$

(due to time scales)

Quasi-Steady State

const.

$$\frac{dP}{dt} = \alpha_p M(t) - \delta_p P$$

$$\Rightarrow \frac{dP}{dt} = \alpha_p M_{ss} - \delta_p P$$

is nearly const.  $M_{ss}$  for the most of the time  $P$  is being produced

$$\int \frac{dP}{\alpha_p M_{ss} - \delta_p P} = \int dt$$

$$P(t) = \frac{\alpha_p M_{ss}}{\delta_p} (1 - e^{-\delta_p t})$$

$$P(t) = \frac{\alpha_p \alpha_m}{\delta_p \delta_m} (1 - e^{-\delta_p t})$$

$$P - \frac{1}{\cancel{\alpha_p M_{ss}} \delta_p} \ln(\alpha_p M_{ss} - \delta_p P) = t + C$$

$$@ t = 0, P = P_0$$

$$C = - \frac{1}{\cancel{\alpha_p M_{ss}} \delta_p} \ln(\alpha_p M_{ss} - \delta_p P_0)$$

$$\Rightarrow - \frac{1}{\cancel{\alpha_p M_{ss}} \delta_p} \ln(\alpha_p M_{ss} - \delta_p P) = t - \frac{1}{\cancel{\alpha_p M_{ss}} \delta_p} \ln(\alpha_p M_{ss} - \delta_p P_0)$$

$$\Rightarrow \frac{1}{\cancel{\alpha_p M_{ss}} \delta_p} \ln\left(\frac{\alpha_p M_{ss} - \delta_p P_0}{\alpha_p M_{ss} - \delta_p P}\right) = t$$

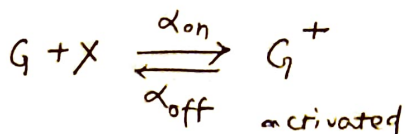
$$\Rightarrow \frac{\alpha_p M_{ss} - \delta_p P}{\alpha_p M_{ss} - \delta_p P_0} = e^{-\delta_p t}$$

Response time is controlled by  $\delta$

$$T_{1/2} = \frac{\ln 2}{\delta}$$

for mRNA  $\delta_m$  is much higher than  $\delta_p$   
 $\delta_m \gg \delta_p$

### Activation of Gene



$$\frac{dG^+}{dt} = \alpha_{on} G X - \alpha_{off} G^+$$

$$\boxed{\alpha_{m+} \gg \alpha_{m0}}$$

Modified transcription rate

$$\frac{dM}{dt} = \underbrace{\alpha_{m0} \cdot G}_{\text{basal rate of expression}} + \alpha_{m+} G^+ - \delta_m M$$

$$\frac{dP}{dt} = \alpha_p M - \delta_p P$$

Constraints

$$G + G^+ = G_{\text{total}}$$

$$G = G_{\text{total}} - G^+$$

Total  
Number  
of Promoter  
fixed

$$\frac{dG^+}{dt} = \alpha_{\text{on}} G X - G^+ \alpha_{\text{off}}$$

$$\Rightarrow \frac{dG^+}{dt} = \alpha_{\text{on}} (G_{\text{t}} - G^+) X - G^+ \alpha_{\text{off}}$$

To solve  $\frac{dM}{dt}$ , we need  $G^+$  only.

Quasi-steady state assumption

$$\frac{dG^+}{dt} = \alpha_{\text{on}} (G_{\text{t}} - G^+) X - G^+ \alpha_{\text{off}} = 0$$

$$\Rightarrow (G_{\text{t}} - G^+) X = G^+ \left( \frac{\alpha_{\text{off}}}{\alpha_{\text{on}}} \right) \rightarrow \text{Dissociation rate } \alpha_d$$

$$\Rightarrow G_{\text{t}} X - G^+ X = G^+ \alpha_d$$

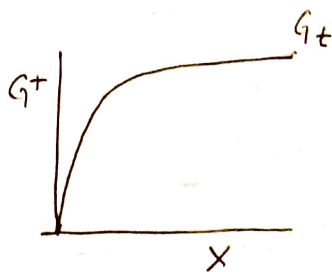
$$\Rightarrow G^+ = \frac{G_{\text{t}} X}{\alpha_d + X}$$

$$\text{Now } \frac{dM}{dt} \approx 0 = \alpha_{\text{mo}} G + \alpha_{\text{mt}} G^+ - \delta_m M$$

for  
simplicity

$$\Rightarrow \frac{dM}{dt} = \alpha_{\text{mt}} \frac{G_{\text{t}} X}{\alpha_d + X} - \delta_m M$$

$$G^+ = G_{\text{t}} \cdot \frac{X}{\alpha_d + X}$$



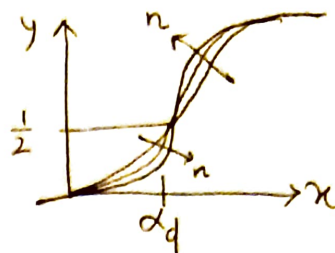
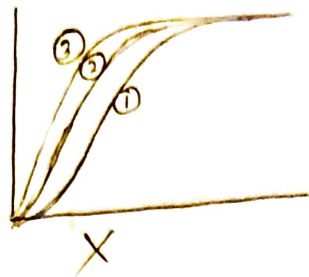
$$y = \frac{x}{\alpha_d + x} \quad (1)$$

$$y = \frac{x^2}{\alpha_d^2 + x^2} \quad (2)$$

$$y = \frac{x^3}{\alpha_d^3 + x^3} \quad (3)$$

$$y = \frac{x^n}{\alpha_d^n + x^n} \quad (n > 1)$$

Hill's function

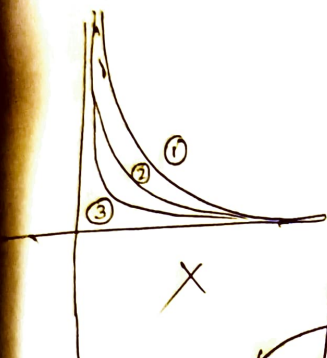


$$y = \frac{\alpha_d}{\alpha_d + x} \quad (1)$$

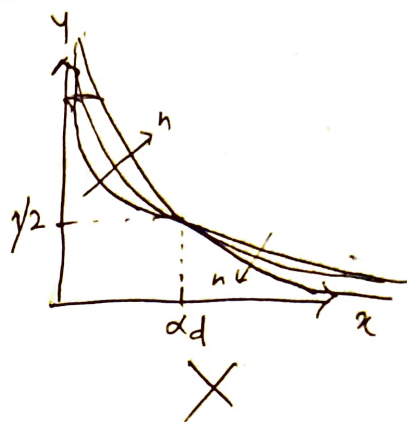
$$y = \frac{\alpha_d^2}{\alpha_d^2 + x^2} \quad (2)$$

$$y = \frac{\alpha_d^3}{\alpha_d^3 + x^3} \quad (3)$$

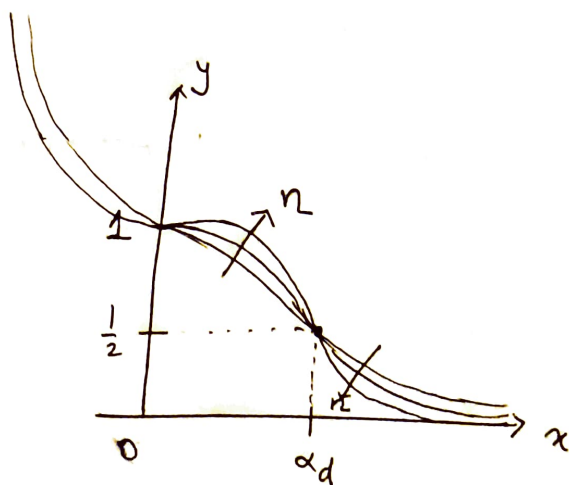
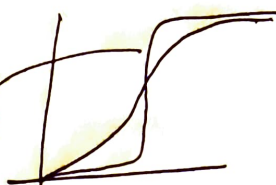
$$y = \frac{\alpha_d^n}{\alpha_d^n + x^n} = \frac{1}{2}$$



@  $x = \alpha_d$   
Co-operativity  
(comes from  $n$ )



Ultra-sensitive



Drug Response

$$\frac{D^n}{EC_{50}^n + D^n}$$

# SS607: Comparative Physiology

25.11.2020

## Epidemiology - Epidemic/Pandemic

→ Studying of Infectious Diseases and their spread

→ Gauging the impact of different vaccination programmes

→ Control/Eradication of Diseases

Simple ODE based Model -

(SIR Model)

Kermack and McKendrick, 1927

→ to understand 'ignoring' birth and death

Spanish Flu (great influenza) - 1920

Ignore

- Age Dynamics

- Geographical Distribution  
to simplify the model

Objective - ① spread of infection. → time dependent variables

Initial population / not infected

or

→ (Susceptible population

→ (Infected population

→ (Recovered population → Re-infected

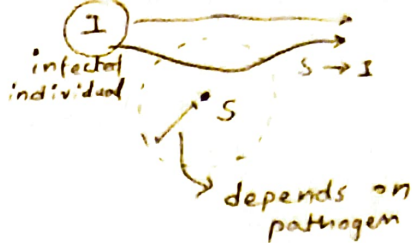
SIRS Model

Basic Assumptions - No. of new infectives in  $\Delta t$  time

$$I(t+\Delta t) - I(t) \propto I$$
$$\propto I$$
$$\propto \Delta t$$

$$I(t+\Delta t) - I(t) \propto S \cdot I \cdot \Delta t$$

new parameter,  $\beta$  infectivity / rate?



Probability of I passing through this region  $P = \beta \Delta t$

probability of S getting infected.

Probability of at least 1 infected individual passing through the region, if total I number of infected people in the population.

Binomial Expansion

$$1 - (1-P)^I = \left\{ 1 - \left( 1 - PI + \underbrace{{}_2I P^2 + \dots}_{\text{very small because } P \text{ very small}} \right) \right\}$$

$$= 1 - 1 + PI$$

$$= PI$$

$$= \beta I \Delta t$$

Assumption, S number of susceptible people.

probability of any S getting infected.

$$= \beta I S \Delta t$$

# if  $P = 0.01$ ,  $I = 5$

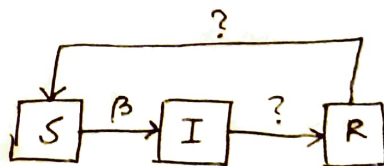
$$1 - (1 - 0.01)^5 = 0.049$$

$$0.01 \times 5 = 0.05$$

$$\frac{I(t+\Delta t) - I(t)}{\Delta t} \propto SI$$

$$\Rightarrow \frac{I(t+\Delta t) - I(t)}{\Delta t} = \beta SI$$

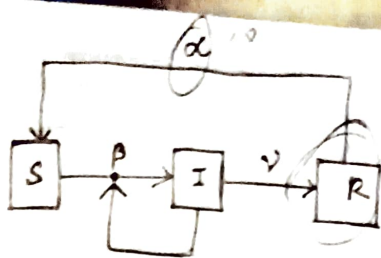
$$\Rightarrow \lim_{\Delta t \rightarrow 0} \frac{dI}{dt} = \beta SI$$



$$\frac{dS}{dt} = ?$$

$$\frac{dI}{dt} = ?$$

$$\frac{dR}{dt} = ?$$



New Parameters :

$\gamma$  = rate of infected to Recovered

$\alpha$  = rate of Recovered to Susceptible

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dS}{dt} = -\beta SI + \alpha R$$

$$\frac{dR}{dt} = \gamma I - \alpha R$$

Total S, I, R conserved

$$N = S + I + R$$

$$R = N - (S + I)$$

$$\frac{d}{dt}(I + S + R) = 0 \Rightarrow \frac{dN}{dt} = 0 \quad (\text{No birth \& Death})$$

Substitute R (Not interested)

$$\left. \begin{aligned} \frac{dS}{dt} &= -\beta SI + \alpha (N - I - S) \\ \frac{dI}{dt} &= \beta SI - \gamma I \end{aligned} \right\}$$

$$\begin{aligned} S + I + R &= N \\ S &= N - I - R \end{aligned}$$

Calculate steady state values of S and I

$$\frac{dI}{dt} = 0$$

$$\frac{dS}{dt} = 0$$

$$I_{ss} = 0 \rightarrow S_{ss} = N$$

$$\Rightarrow \beta S_{ss} I_{ss} - \gamma I_{ss} = 0 \quad \text{--- (I)}$$

$$\bullet \beta S_{ss} I = \alpha (N - I - S_{ss}) \quad \text{--- (II)}$$

$$\Rightarrow \beta S_{ss} = \gamma$$

$$\Rightarrow \beta \frac{\gamma}{\beta} I = \alpha (N - I - \frac{\gamma}{\beta})$$

$$\Rightarrow \underline{S_{ss} = \frac{\gamma}{\beta}}$$

$$\Rightarrow \gamma I_{ss} = \alpha N - \alpha I_{ss} - \frac{\alpha \gamma}{\beta}$$

$$\Rightarrow I_{ss}(\gamma + \alpha) = \alpha N - \frac{\alpha \gamma}{\beta}$$

$$\underline{I_{ss} = \frac{\alpha N - \frac{\alpha \gamma}{\beta}}{\gamma + \alpha}}$$

(I) + (II)

$$\begin{aligned} \alpha N - \alpha S - \alpha I - \gamma I &= 0 \\ \alpha N - \alpha S - I(\alpha + \gamma) &= 0 \\ I &= \frac{\alpha(N - S)}{\alpha + \gamma} \end{aligned}$$

Steady state 1

$$S_{ss} = N$$

$$I_{ss} = 0$$

Steady state 2

$$S_{ss} = \frac{\nu}{\beta}$$

$$I_{ss} = \frac{\alpha(N - \frac{\nu}{\beta})}{\alpha + \nu}$$

$$N - \frac{\nu}{\beta} = 0$$

$$\Rightarrow \nu = \beta N$$

If  $N < \frac{\nu}{\beta}$  or  $\boxed{\frac{N\beta}{\nu} < 1}$  system becomes unstable

$R_0$

(Reproduction Number)

Infection never becomes a pandemic

$R_0 \rightarrow$  expected number of infected by a single infected individual.

Read about : correlation of mortality of the diff. pathogens, with the  $R_0$  values

$$S = S_0 e^{-\beta N}$$

= ode45 (function, tspan, y0)

$$\frac{dS}{dt} = -\beta SI$$

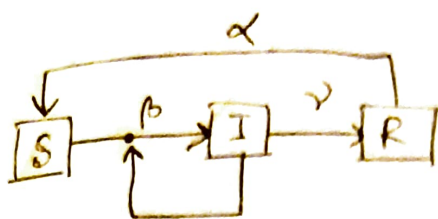
$$\frac{dI}{dt} = \beta SI - \nu I$$

$$I_{ss} = 0$$

$$S_{ss} =$$

$$S_{ss} = \frac{\nu}{\beta}$$





$\beta$   
 $\gamma$   
 $\alpha$

$$\frac{dS}{dt} = -\beta SI + \alpha R$$

$$\frac{dI}{dt} = +\beta SI - \alpha \gamma I$$

$$\frac{dR}{dt} = \gamma I - \alpha R$$

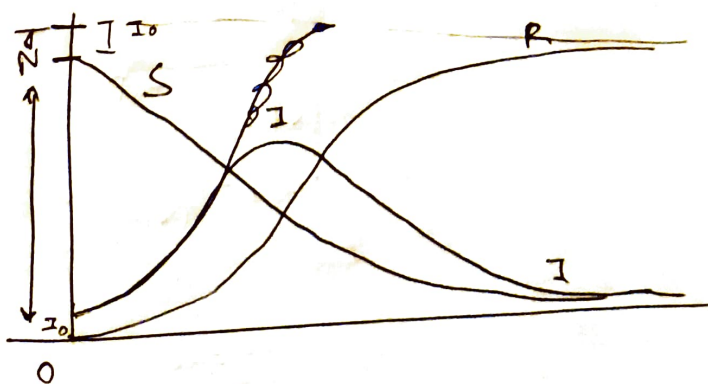
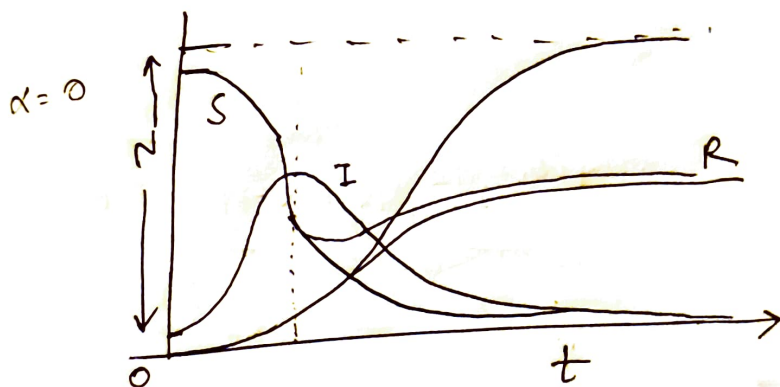
$$N = S + I + R$$

$$\frac{N\beta}{\gamma}$$

$$-\beta SI + \alpha(N - S - I)$$

$$\beta SI - \gamma I$$

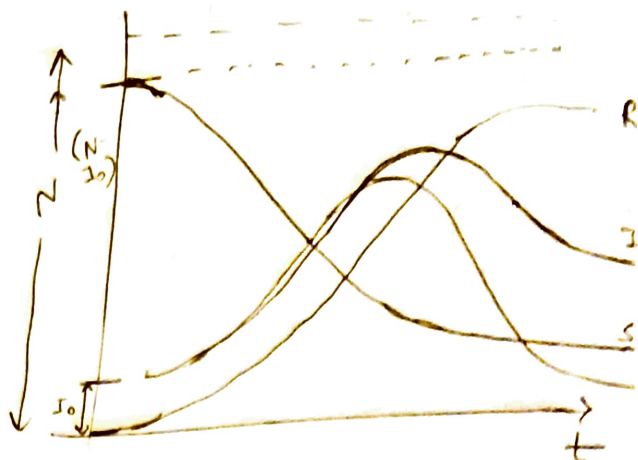
Steady state.



$$\frac{dS}{dt} = -\beta SI + \alpha R$$

$$\frac{dI}{dt} = \beta SI - \nu I$$

$$\frac{dR}{dt} = \nu I - \alpha R$$



$\alpha \neq 0$



$$\begin{aligned} \nu &= 0.12 \\ \alpha &= 0.002 \\ \beta &= 5 \times 10^{-9} \end{aligned}$$

$$N = 6 \times 10^6$$

$$\frac{\alpha(N - \frac{\nu}{\beta})}{\alpha + \nu}$$

$$\frac{N'\beta}{\nu} < 1$$

$$N' = N - pN = N(1-p)$$

↓  
% of pop<sup>n</sup> vaccinating.

~~3~~

$$N' = \frac{\nu}{\beta}$$

$$\frac{\nu}{\beta} = N(1-p)$$

$$\Rightarrow \frac{\nu}{N\beta} = 1-p$$

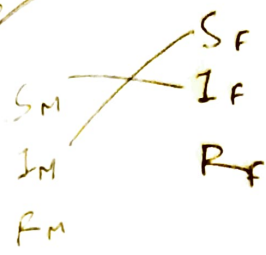
$$\Rightarrow p = \left(1 - \left(\frac{\nu}{N\beta}\right)\right) \times 100\%$$

$$\frac{\nu}{N\beta}$$

$$p = \left(1 - \frac{1}{R_0}\right)$$

$$= \left(1 - \frac{1}{R_0}\right) \times 100\%$$

STD / STI

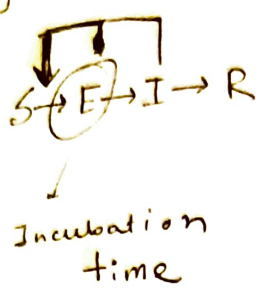
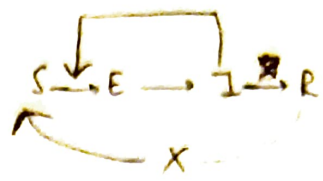


$$\frac{dI_M}{dt} = \beta I_M S_{\text{of}}$$

$$\frac{dI_F}{dt} = \beta I_F S_M$$



Latency, Incubation Time.



$$\frac{dE}{dt} = \beta S I - \epsilon E$$

$$\frac{dI}{dt} = \epsilon E - \nu I$$

Steady State

$$N = S + E + I + R$$

$$E = N - S - I - R$$

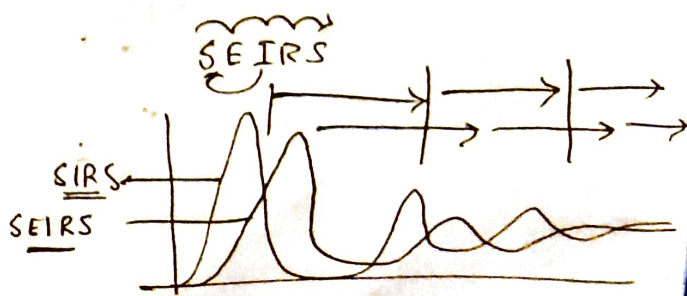
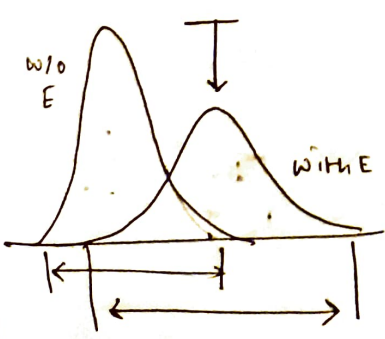
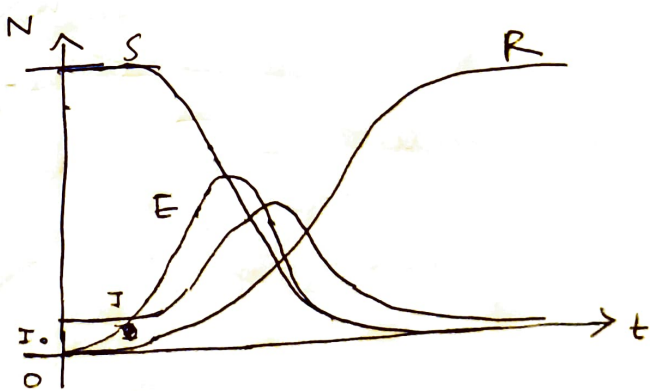
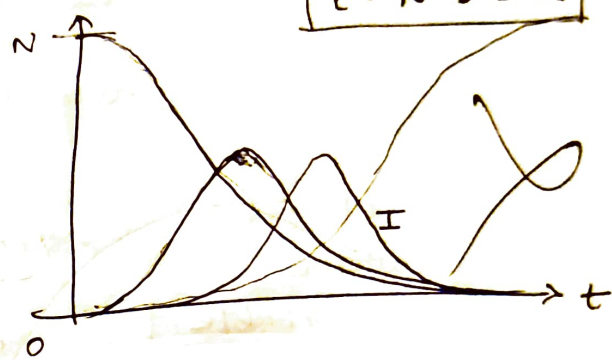
$$\frac{dS}{dt} = -\beta S I + \alpha R = 0$$

$$\frac{dE}{dt} = +\beta S I - \epsilon E = 0$$

$$\frac{dI}{dt} = \epsilon E - \nu I = 0$$

$$\frac{dR}{dt} = \nu I - \alpha R = 0$$

$$I = \frac{\epsilon E}{\nu}$$



~~23/03/2026~~

## FB667 : Computational Physiology

23/03/2026

### "Stability Analysis" of the SIRS Model

Non-linear

$$\begin{cases} \frac{dS}{dt} = -\beta SI + \delta(N-S-I) = f_1 \\ \frac{dI}{dt} = \beta SI - \nu I = f_2 \end{cases} \quad R = N - S - I$$

Linearised version

Jacobian

$$\frac{d\bar{U}}{dt} = J \cdot \bar{U} \rightarrow \text{vector } \bar{U} = \begin{bmatrix} \Delta S(t) \\ \Delta I(t) \end{bmatrix}$$

$$U = A_1 e^{\lambda_1 t} v_1 + A_2 e^{\lambda_2 t} v_2$$

Local stability

$$\underline{\text{Re}(\lambda_1) < 0 \rightarrow \text{stable}}$$

$$\underline{\text{Re}(\lambda_1) > 0 \rightarrow \text{unstable}}$$

$$\begin{array}{cc} \text{SS1} & \text{SS2} \\ \{N, 0\} & \left\{ \frac{\nu}{\beta}, \frac{\delta}{\nu} \right\} \end{array}$$

$$\left\{ \frac{\nu}{\beta}, \frac{\frac{\nu}{\beta}(\frac{\nu}{\beta}-1)}{\nu+\alpha} \right\}$$

$\lambda_1, \lambda_2 \rightarrow \text{eigenvalues}$

$v_1, v_2 \rightarrow \text{eigen vectors}$

Calculation of Jacobian  $\rightarrow$   
Matrix

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial S} & \frac{\partial f_1}{\partial I} \\ \frac{\partial f_2}{\partial S} & \frac{\partial f_2}{\partial I} \end{bmatrix}$$

using  $\frac{d\bar{U}}{dt} = J \cdot \bar{U}$

$$= \begin{bmatrix} -\beta I - \delta & -\beta S - \delta \\ \beta I & \beta S - \nu \end{bmatrix}$$

$$\frac{d}{dt}(\Delta S) = (-\beta I - \delta) \Delta S(t) + (-\beta S - \delta) \Delta I(t) \quad \text{--- (I)}$$

$$\frac{d}{dt}(\Delta I) = \beta I \Delta S(t) + (\beta S - \nu) \Delta I(t) \quad \text{--- (II)}$$

Linear equations.

Calculate the Eigen values for Jacobian Matrix.

$$|J - \lambda I| = 0$$

$$J = \begin{bmatrix} -\beta I^* - \delta & -\beta S^* - \delta \\ \beta I^* & \beta S^* - \gamma \end{bmatrix}$$

$S^*$   
 $I^*$  Steady States.

SS1 = Disease Free Equilibrium (DFE)

$\{N, 0\}$

When,  $R_0 < 1$

$$J = \begin{bmatrix} -\beta \times 0 - \delta & -\beta N - \delta \\ -\beta \times 0 & \beta N - \gamma \end{bmatrix} = \begin{bmatrix} -\delta & -\beta N - \delta \\ 0 & \beta N - \gamma \end{bmatrix}$$

$$(\lambda I - J) = \begin{bmatrix} \lambda + \delta & \beta N + \delta \\ 0 & \lambda - \beta N + \gamma \end{bmatrix}$$

$$\begin{vmatrix} \lambda + \delta & \beta N + \delta \\ 0 & \lambda - \beta N + \gamma \end{vmatrix} = 0$$

$$\Rightarrow (\lambda + \delta)(\lambda - \beta N + \gamma) = 0$$

|                                |
|--------------------------------|
| $\lambda_1 = -\delta$          |
| $\lambda_2 = \beta N - \gamma$ |

$$\text{Re}(\lambda_1) < 0 \rightarrow \text{Unstable}$$

$$= \gamma \left( \frac{\beta N}{\gamma} - 1 \right)$$

$$= \gamma (R_0 - 1) \quad \text{if } R_0 < 1$$

$\therefore$  SS1 is stable if  $R_0$  is  $< 1$   
SS1 is unstable if  $R_0$  is  $> 1$

552

$$\left\{ \frac{\gamma}{\beta}, \frac{\alpha \frac{\gamma}{\beta} \left( \frac{N\beta}{\gamma} - 1 \right)}{\alpha + \gamma} \right\}$$

Endemic Equilibrium

$$J = \begin{bmatrix} -\beta \cdot \frac{\alpha \gamma \left( \frac{N\beta}{\gamma} - 1 \right)}{\alpha + \gamma} - \delta & -\beta \cdot \frac{\gamma}{\beta} - \delta \\ \beta \cdot \frac{\alpha \gamma \left( \frac{N\beta}{\gamma} - 1 \right)}{\alpha + \gamma} & \beta \cdot \frac{\gamma}{\beta} - \gamma \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\alpha \gamma \left( \frac{N\beta}{\gamma} - 1 \right)}{\alpha + \gamma} - \delta & -\gamma - \delta \\ \frac{\alpha \gamma \left( \frac{N\beta}{\gamma} - 1 \right)}{\alpha + \gamma} & 0 \end{bmatrix}$$

$$|\lambda I - J| = 0$$

Routh Hurwitz Theorem.

If  $\text{Trace}(J) < 0$   
 $|J| > 0$

Then eigen values are -ve.

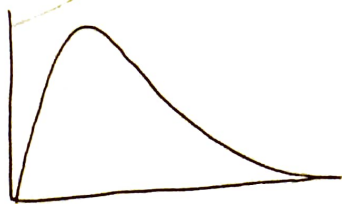
Applicable for all  $(n \times n)$  matrix.

$$|J| = -\beta I^* \cdot (-\gamma - \delta)$$

$$\therefore \beta I^* (\gamma + \delta) > 0 \quad \text{if } R_0 > 0$$

$$\text{Trace}(J) = (-\beta I^* - \delta + 0) < 0$$

Rayleigh Distribution.



# GB667: Computational Physiology

26/03/2021

SIRS - Stochastic

"Gillespie Algorithm"

likelihood  
↑  
propensity / rate of reaction

# well mixed system

Species  $S_1 \dots S_N$  and rxn  $R_1 \dots R_M$

State at  $t$

amount of  $S_i$  @  $t$

$$X(t) = \{x_1(t), x_2(t), \dots, x_N(t)\}$$

for a reaction

$R_j$  Stoichiometric change vector  $v_j$

$$X \leftarrow X + v_j$$

Zeroth order reaction  $\emptyset \xrightarrow{R_1, c_j} S_i \rightarrow a_j(x) = c_j$   
(const.)

First order reaction?  $S_i \xrightarrow{R_2, c_j} \emptyset \rightarrow a_j(x) = c_j x_i(t)$

$R_3$   
 $S_i + S_k \xrightarrow{c} \rightarrow a_j(x) = x_i(t) x_k(t) c$

Master Equation

$$P = C e^{-a_0 C}$$

$$\frac{d}{dt} P(x,t) = \sum_{j=1}^m \left[ a_j(x-v_j) \overbrace{P(x-v_j,t)}^{\text{Probability}} - \underbrace{a_j(x) P(x,t)}_{\text{Probability}} \right]$$

add all the  $m$  propensity.

$$a_i = a_j(x(t))$$

$$a_0 = \sum_{j=1}^m a_j$$

$\frac{a_j}{a_0} \rightarrow$  likelihood of  $j^{\text{th}}$  individual reaction happening  $\rightarrow$  Probability

generate two random variable  $r_1, r_2$  using uniform distribution  $[0, 1]$

time constant

$$\tau = \frac{1}{a_0} \ln\left(\frac{1}{r_1}\right)$$

if for example  $(a_j \geq r_2 a_0)$

then

$$x_j = x_j + v_j \quad \text{update state}$$

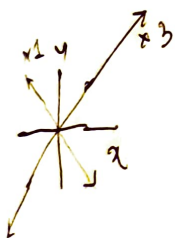
$\tau$  plot : stochastic time gaps.

Time to next reaction.

why  $r_1, r_2$  two random variables.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$|\lambda I - A| = 0 \quad \begin{vmatrix} \lambda - 2 & -1 \\ -1 & \lambda - 2 \end{vmatrix} = 0$$



$$(\lambda - 2 - 1)(\lambda - 2 + 1) = 0 \Rightarrow (\lambda - 2)^2 - 1 = 0$$

$$\Rightarrow \lambda_1 = 3, \lambda_2 = 1$$

$$\Rightarrow \lambda^2 - 2 \cdot 2 \cdot \lambda + 4 - 1 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - \lambda + 3 = 0$$

$$\Rightarrow \lambda(\lambda - 3) - 1(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 3) = 0 \Rightarrow \boxed{\lambda = 1, 3}$$

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$2x + y = x \quad \text{eigenvector } \lambda_2$$

$$\Rightarrow x + y = 0$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\Rightarrow$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$2x + y = 3x \Rightarrow y - x = 0 \Rightarrow x = y$$

$$x + 2y = 3y \Rightarrow$$

$\lambda_1$  eigenvector

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$X =$

# PCA Principal Component Analysis

$$\begin{matrix} \text{Patients} \\ \left[ \begin{array}{cccc} 1 & \dots & \dots & d \\ \vdots & & & \\ n & & & \end{array} \right] \end{matrix}$$

Features

Calculate a Covariance Matrix

$$\Sigma = \frac{1}{n} X^T X$$

Covariance  
correlation coefficient

↓  
eigen values  
and  
eigen vectors

$$\Sigma \cdot U = \lambda \cdot U$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}_{3 \times 2}$$

$$X^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}_{2 \times 3}$$

$$\frac{1}{n} X^T X = \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2}$$

$$\lambda = \frac{\lambda_1}{3}, \frac{\lambda_2}{3}$$

$$= \frac{1}{3}, 1$$

$$= \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}_{2 \times 2} \dots \dots \lambda_1 + \lambda_2$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\dot{x} = \overset{\text{Matrix}}{(A)} x(t)$$

$$\Rightarrow x(t) = ( ) e^{At}$$

replace with Taylor series.

Diagonalization.

$$A = V \Lambda V^{-1}$$

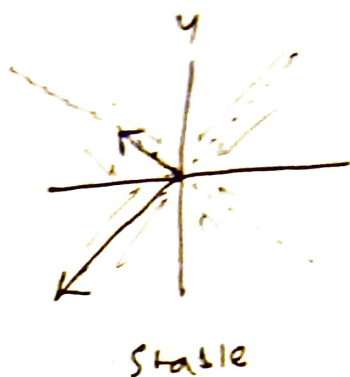
$$\Lambda = \text{diagonal}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

$V$  = eigen vector matrix

$$x(t) = \underset{\substack{\uparrow \\ \text{eigen} \\ \text{vector}}}{V} e^{\overset{\substack{\uparrow \\ \text{eigen values}}}{\Lambda} t} \underset{\text{initial condition}}{(x_0)}$$

$$A = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

$$\begin{array}{cc} \lambda_1 & \lambda_2 \\ -1 & -3 \\ \begin{bmatrix} 1 \\ -1 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{array}$$

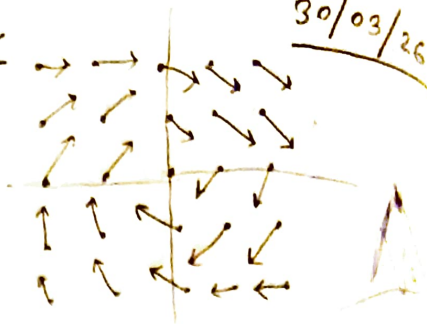
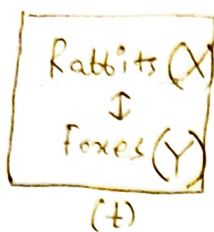


stable

# BB667 : Computational Physiology

30/03/26

## Phase Portrait / Diagram



How this system evolves with time

$$X(t)$$

$$Y(t)$$

Framework  
ODE

$$\frac{dx}{dt}, \frac{dy}{dt}$$

$$\frac{dx}{dt} = +\alpha X - \gamma XY$$

$$\frac{dy}{dt} = 0 + \beta XY - \delta Y$$

$$\frac{dx}{dt} = +\alpha X - \gamma XY$$

$$\frac{dy}{dt} = +\beta XY - \delta Y$$

$$\# \quad \frac{dx}{dt} = 2\lambda - x\gamma \quad \frac{dy}{dt} = x\gamma - \gamma$$

Phase Portrait

Calculate Nullclines

$$X \text{ Nullcline } \frac{dx}{dt} = 0$$

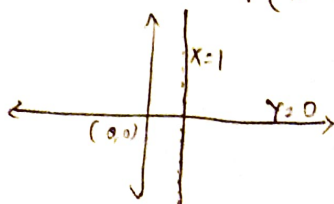
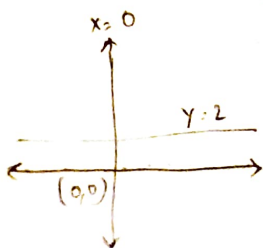
$$Y \text{ Nullcline } \frac{dy}{dt} = 0$$

$$\frac{dx}{dt} = [2\lambda - x\gamma] = 0$$

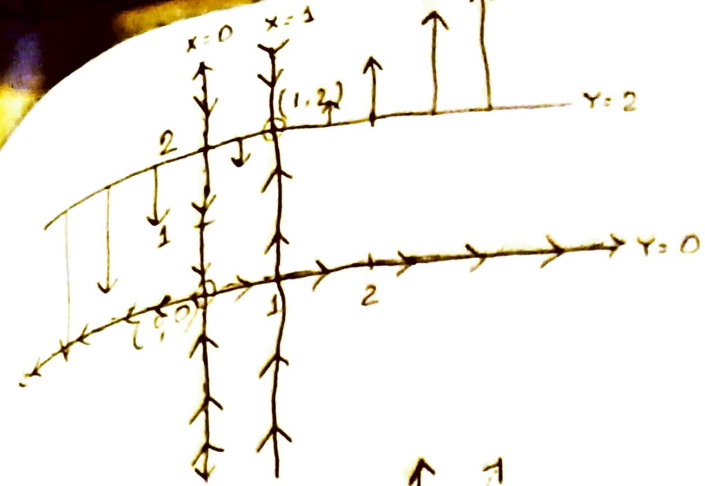
$$(2-\gamma)x = 0$$

$$\frac{dy}{dt} = x\gamma - \gamma = 0$$

$$\gamma(x-1) = 0$$



Intersections  $\rightarrow$  steady states



|                                   |                                  |
|-----------------------------------|----------------------------------|
| $\frac{dx}{dt} = +ve \rightarrow$ | $\frac{dy}{dt} = +ve \uparrow$   |
| $\frac{dx}{dt} = +ve \rightarrow$ | $\frac{dy}{dt} = +ve \uparrow$   |
| $\frac{dx}{dt} = -ve \leftarrow$  | $\frac{dy}{dt} = -ve \downarrow$ |
| $\frac{dx}{dt} = -ve \leftarrow$  | $\frac{dy}{dt} = -ve \downarrow$ |



for X Nullcline calculate  $\frac{dy}{dt}$   
for Y Nullcline calculate  $\frac{dx}{dt}$

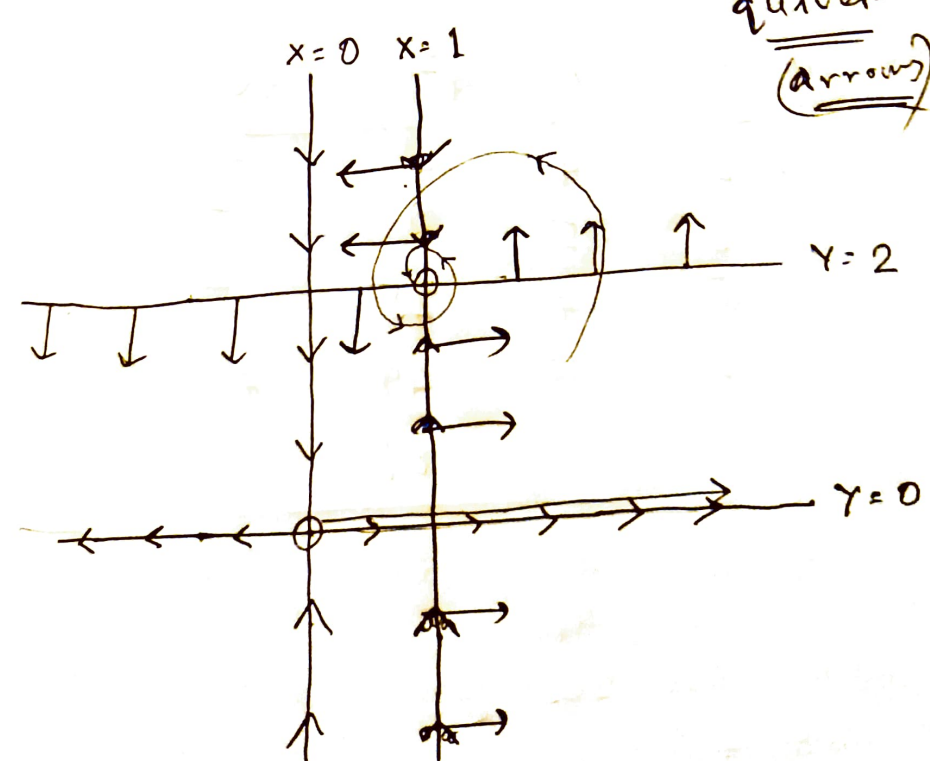
for  $x=0$   
 $\frac{dy}{dt} = Y(0-1) = -Y$

$x=1$   
 $\frac{dx}{dt} = (2-y)x$   
 $= (2-y)$

$y=0$   
 $\frac{dx}{dt} = (2-0)x$   
 $= 2x$

$y=2$   
 $\frac{dy}{dt} = 2(x-1) = 2x-2$

quiver  
(arrows)



$$\frac{dx}{dt} = 2x - xy = 0$$

$$\frac{dy}{dt} = xy = 0$$

$$x(2-y) = 0$$

$$x = 0 \longrightarrow \frac{dx}{dt} = 0$$

$$y = 0 \longrightarrow \frac{dx}{dt} = 2x$$

$$\frac{dy}{dt} = 0 \longleftarrow \begin{matrix} x = 0 \\ y = 2 \end{matrix}$$

$$\frac{dy}{dt} = 2x$$

